

|   |     |      |                                                                                                                   |            |
|---|-----|------|-------------------------------------------------------------------------------------------------------------------|------------|
|   |     |      | hence rate of change of magnetic flux is greater. This induced e.m.f. is greater.                                 |            |
| 7 | (a) | (i)  | effective direct voltage that produces the same heating effect as the alternating voltage.                        | 1          |
|   |     | (ii) | $V_o = 325.269 \text{ V}$<br>$\text{Peak P} = 325.269 \times 0.26\sqrt{2} = 120 \text{ W}$                        | 1<br>1     |
|   | (b) |      | $\langle V^2 \rangle = \frac{(10)^2(2.0) + (5^2)(1.0)}{4.0}$ $= 56.25 \text{ V}$ $V_{\text{rms}} = 7.5 \text{ V}$ | 1<br><br>1 |

**30**

**Section B**

|          |            |                                         |  |
|----------|------------|-----------------------------------------|--|
| <b>9</b> | <b>(a)</b> | Calculate speed for $d = 10, 20, 30$ m: |  |
|----------|------------|-----------------------------------------|--|